Pankaj Baid

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LinkedIn • Github • Leetcode • Personal Portfolio

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2024-2028	Bachelor of Technology (Computer Science and AI)	Newton School of Technology, Rishihood University	8.618/10.0
2022-2023	Intermediate (Class XII)	DAV RB KEDIA	86.0%
2020-2021	Matriculation (Class X)	DAV RB KEDIA	92.33%

EXPERIENCE

Founding Software Engineer (AI & Backend) September 2024 - March 2026
in.culcate

- Built **20+ REST** endpoints in Node.js, Express.js, MongoDB for auth and content in a **2,000-user app**.
- Engineered a **5+ agent pipeline** (scripting, voice, ElevenLabs narration), cutting production time **40%**.
- Automated **5+ publishing workflows** with a centralized CMS, reducing content turnaround time by 30%.
- Streamlined the team's deployment workflow, running backend services on **AWS EC2**.

PROJECTS

VITRAM – Financial companion for salaried users in India (Github) (Demo) October 2026

- Optimised via profiling, caching and indexing: dashboard throughput 3.5x, hot query 4.7 ms to 0.25 ms.
- Architected a 129-operation FastAPI REST API over 38 PostgreSQL tables, consumed by a React UI.
- Deployed on free-tier Cloudflare Workers, Render, Neon; k6: 213 req/s/process, p95 151 ms (synthetic).
- Hardened with 656 tests (88% coverage) incl. Hypothesis property tests; EMI mutation score 89.3%.

Resilient TaskFlow – Crash-safe distributed job queue (Github) (Demo) September 2026

- Designed outbox and fenced leases on PostgreSQL + Redis Streams: 0 of 10,000 jobs lost on SIGKILL.
- Benchmarked with k6 on one laptop: 500 submissions/s, p99 346 ms, 0 errors (one API process).
- Implemented 14 REST endpoints on Node.js, Fastify, TypeScript with JWT auth and idempotency keys.
- Verified with 300+ tests, incl. 46 Docker integration tests with real workers, in GitHub Actions CI.

ACHIEVEMENTS

- Team rank 8 on the public leaderboard among ~80,000 registered participants, Amazon ML Challenge 2026.
- Merged a fix to Hugging Face Transformers (168k-star repo, PR #42373): TensorDict padding regression.
- CodeChef 3-star (rating 1638); 600+ DSA problems solved across platforms.
- Merged 2 **PostgreSQL RLS** fixes in **AOSSIE Ell-ena** (PRs #76, #78): EXISTS policies, missing DELETE rules.
- ;Winner at LMNIT-AOSSIE Unstoppable Hackathon; 2nd place in the app-development hackathon at IIT Roorkee Cognizance out of 500 participants.

SKILLS

- **Databases & Data Tools:** MySQL, MongoDB, PostgreSQL
- **Frameworks & Libraries:** Tailwind, Next JS, React Native, Prisma ORM, React, Redux, Node.js, Express JS, FastAPI
- **Computer Language:** Python, JavaScript, C++, TypeScript, SQL
- **Generative AI & LLMs:** RAG, LangChain, LangGraph, LLM, GenAI, Hugging Face

RELEVANT COURSEWORK

Data Structures & Algorithms; System Design; Database Management Systems (DBMS); Computer Networks;